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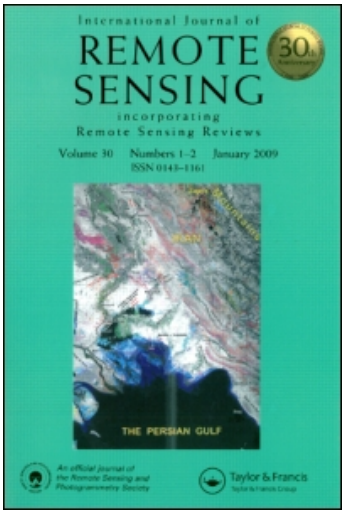

International Journal of Remote Sensing
Publication details, including instructions for authors and subscription information:
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Online publication date: 22 March 2011

To cite this Article Xu, Xiaojun , Du, Huaqiang , Zhou, Guomo , Ge, Hongli , Shi, Yongjun , Zhou, Yufeng , Fan, Weiliang and Fan, Wenyi(2011) 'Estimation of aboveground carbon stock of *Moso* bamboo (*Phyllostachys heterocycla* var. *pubescens)* forest with a Landsat Thematic Mapper image', International Journal of Remote Sensing, 32: 5, 1431 — 1448
To link to this Article: DOI: 10.1080/01431160903551389
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International Journal of Remote Sensing
Vol. 32, No. 5, 10 March 2011, 1431–1448

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Estimation of aboveground carbon stock of *Moso* bamboo (*Phyllostachys heterocycla* var. *pubescens)* forest with a Landsat Thematic Mapper image

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(*Received 24 June 2009; in final form 8 December 2009*)

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The extensive distribution of bamboo forests in South and Southeast Asia plays an important role in the global carbon budget. It is an urgent task to accurately and in good time estimate carbon stock within these areas. In this study, linear regression, partial least-squares (PLS) regression and backpropagation artificial neural network (BP-ANN) with a Gaussian error function as the activation function of the hidden layers (Erf-BP) were used to estimate aboveground carbon (AGC) stock of *Moso* bamboo in Anji, Zhejiang Province, China. Based on the combined use of Landsat Thematic Mapper (TM) and field measurements, the results indicate that the Erf-BP model provided the best estimation performance, and the linear regression model performed the poorest. This study indicates that remote sensing is an effective way of estimating AGC of *Moso* bamboo in a large area.

1. Introduction

Bamboo forests are distributed extensively in South and Southeast Asia and are increasing continuously (Sharma 1980). China has a bamboo forest area of approximately 54 400 km^2, in which *Moso* bamboo accounts for approximately 70% (Lobovikov *et al.* 2006). Because of its rapid growth rate and strong capacity for absorbing and storing carbon in forests and products, bamboo plays an important role in both the regional and global carbon cycle (Zhou 2006). A bamboo shoot can reach crown height within a few months, and culms are cut every 2 years for production. Vigorous growth makes bamboo a particularly attractive plant for carbon sequestration (Lobovikov *et al.* 2006). Under the Clean Development Mechanism of the Kyoto Protocol, a study has been conducted on the role of bamboo in poverty alleviation, carbon sequestration, and climate change (Lobovikov *et al.* 2006). The theme of the VIII World Bamboo Congress in 2009 was 'Bamboo, The Environment, and Climate Change'. Because of its versatility and usefulness, bamboo will continue to play an

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International Journal of Remote Sensing
ISSN 0143-1161 print/ISSN 1366-5901 online © 2011 Taylor & Francis
http://www.tandf.co.uk/journals
DOI: 10.1080/01431160903551389

important role in the world. To examine the role of bamboo forests in Anji, especially *Moso* bamboo, and more importantly global carbon sequestration, it is an urgent task to accurately and in good time estimate bamboo carbon stock.

In the past, many methods and experiments for bamboo biomass estimation have been used (Hunter and Wu 2002). In practice, traditional forest measurement is still the common approach for estimating the aboveground biomass (AGB) and net primary production of bamboo forests (Uchimura 1980, Qiu *et al.* 1992, Shanmughavel and Francis 1996, Isagi *et al.* 1997). This methodology involves people going out into the field and investing a considerable amount of time and labour. Not only is this practice extremely costly in terms of time and labour, it is also difficult to update spatial distribution in a large area. However, remote sensing provides a more efficient way to estimate AGB in large areas, and has already been extensively explored in tropical forests in Brazil, Malaysia and Thailand (Sader *et al.* 1989, Lee and Nakane 1997, Foody *et al.* 2003, Zheng *et al.* 2004, Lu 2005, Lu *et al.* 2005, Hall *et al.* 2006, Myeong *et al.* 2006). However, remote sensing research has rarely been used for estimating the aboveground carbon (AGC) stock for bamboo forests. The challenge has been to identify suitable variables and algorithms to establish the biomass/carbon estimation model (Lu 2006). Some potential methods, such as stepwise regression analysis, correlation analysis and the partial least-squares (PLS)-Bootstrap algorithm, may be used to identify suitable variables.

Linear regression analysis may be the most common method in estimating AGC (Sader *et al.* 1989, Lu *et al.* 2004, Zheng *et al.* 2004, Lu and Batistella 2005, Hall *et al.* 2006). The problem with this method is that the accuracy can be poor (Hyyppä *et al.* 2000, Boyd *et al.* 2002, Foody *et al.* 2003, Muukkonen and Heiskanen 2005, Labrecque *et al.* 2006). Comparing linear regression analysis to PLS regression, PLS regression does not require the assumption of zero error in the predictor data (Curran and Hay 1986). When there are more independent variables (such as vegetation indices and textures) than samples, PLS regression is effective. The backpropagation artificial neural network (BP-ANN) algorithm is the most common algorithm used in ANNs (Zupan and Gasteiger 1991). The BP-ANN can be used to solve complex non-linear problems. Previous research has shown that PLS regression and BP-ANN are superior to the traditional regression analysis approach (Martens and Naes 1989, Bishop 1995, Ham and Kostanic 2000, Hyyppä *et al.* 2000, Boyd *et al.* 2002, Foody *et al.* 2003, Muukkonen and Heiskanen 2005).

Although many methods have been tested for estimating and mapping certain forest AGC, their applicability in bamboo forests is unknown. The primary objective of this study was therefore to identify suitable methods to formulate a reliable way for estimating AGC for *Moso* bamboo by using a two-step process that (1) uses linear regression analysis, PLS and BP-ANN with a Gaussian error function as the activation function of hidden layers (Erf-BP) to estimate and map AGC of *Moso* bamboo, and (2) compares the AGC prediction accuracy of the above three methods. A further objective of this study was to assess the performance of the three methods in estimating AGC of *Moso* bamboo forests.

2. A brief description of the algorithms

2.1 *PLS regression*

The PLS regression method provides a way to reduce a large number of collinear variables into a limited number of uncorrelated, latent structures or components (Norgaard *et al.* 2000, Wolter *et al.* 2008). When sampled, the PLS regression assumes

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the vectors in the predictor space (irrespective of error) provide superior predictive ability for additional observations when there is a high degree of correlation among predictor variables (SAS Institute Inc. 2000). A detailed description of the PLS algorithm is provided in Ehsani *et al.* (1999).

One crucial factor in using the PLS regression is to determine the number of components, and this is often identified by cross-validation (Geisser 1974, Stone 1974, Wold *et al.* 1984). In this study, a minimum sum of the mean relative errors (MREs) of the training and validation sets is proposed to determine the optimal number of PLS components, as expressed as equations (1) and (2). The sum of the MREs of the training and validation sets as a function of the number of PLS components is then used to determine the optimal number of PLS components, that is the number of PLS components that produces a minimum sum of MREs.

$$\mathrm{MRE}_h = \min(\mathrm{MRE}_{\mathrm{Tra},i} + \mathrm{MRE}_{\mathrm{Val},i}) \quad (i = 1, 2, \ldots, p) \tag{1}$$

$$\mathrm{MRE}_{\mathrm{Tra},i} \text{ or } \mathrm{MRE}_{\mathrm{Val},i} = \frac{1}{n} \times \sum_{i=1}^{n} \left| \frac{Y_i - \hat{Y}_i}{Y_i} \times 100\% \right| \tag{2}$$

where $\mathrm{MRE}_{\mathrm{Tra},i}$ and $\mathrm{MRE}_{\mathrm{Val},i}$ are the MREs of the training set and the validation set for the number of components i, respectively; Y_i and $\hat{Y}_i$ are the original data and the predicted values, respectively; n is the number of sample plots; h is the number of optimal components used for establishing an ideal model; and p is the total number of components. This method takes into account the prediction accuracies of both the training and validation sets.

2.2 *BP-ANN*

BP-ANN behaviour is mainly determined by the activation functions of its neurons (Pan *et al.* 2007). Changing activation functions is one way to improve the BP-ANN (Kenue 1991). Xu *et al.* (in press) proposed an Erf-BP algorithm whose activation functions of the hidden and output layers are a Gaussian error function (Erf) and a logistic function, respectively. The experimental results showed that the performance, including convergence speed and accuracy, of the Erf-BP algorithm was slightly better than the standard BP-ANN algorithm whose activation functions of the hidden and output layers are both logistic functions (Xu *et al.* in press). In this study, the Erf-BP algorithm was used to estimate the AGC of *Moso* bamboo. The Erf function and its first-order derivative are expressed by equations (3) and (4) and plotted in figure 1.

$$f(x) = \mathrm{Erf}\left(\frac{x}{\sqrt{\lambda}}\right) = \frac{2}{\sqrt{\pi}} \int_0^{\frac{x}{\sqrt{\lambda}}} e^{-t^2} \mathrm{d}t \tag{3}$$

$$\frac{\mathrm{d}f(x)}{\mathrm{d}x} = \frac{2}{\sqrt{\pi\lambda}} \exp\left(-\frac{x^2}{\lambda}\right) \tag{4}$$

where t is variable, $t \in [0, x/\sqrt{\lambda}]$, and λ is the slope parameter that is used to control the shapes of the Erf plots. Figure 1 shows the Erf function and its first-order derivative for two different values of the slope parameter, $\lambda = 0.5$ and $\lambda = 2$. As λ decreases, the Erf approaches the hard limiter function, and the maximum of the first-order derivative function increases.

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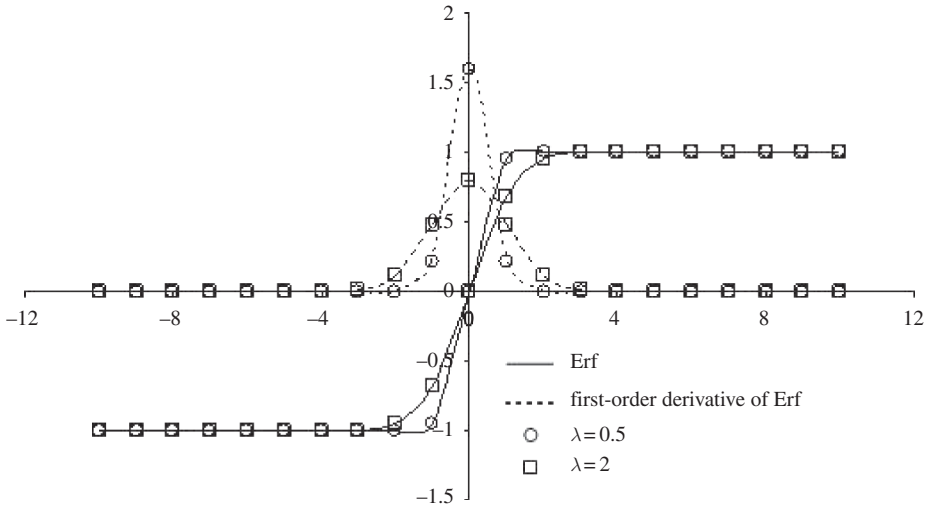

Figure 1. Plots of the Gaussian error function (Erf) and its first-order derivative. The X and Y axes are the input values and output values of the functions, respectively.

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3. Methods

3.1 *Study area*

Anji county is located in the northwestern part of Zhejiang Province, China and has been named the home of *Moso* bamboo forest because of its large area distribution as well as its important role in supporting the local economy. For these particular reasons, Anji county was selected as the study area (see figure 2). The study area covers 1886.45 km^2, of which 71.1% is forested; *Moso* bamboo forest accounts for 45.5% of the forested area. The terrain is undulating with elevation ranging from 4 to 1587 m above sea level. The annual precipitation average is between 1100 and 1900 mm, and annual average temperature is between 12.2°C and 15.6°C.

3.2 *Field inventory and analysis*

A field inventory was conducted during August and September 2008. According to a stratified random sampling scheme, a total of 54 sample plots measuring 0.09 ha (30 m × 30 m) were allocated. Towns within the entire county were used as strata and the percentage of bamboo area of every town as the weight of every stratum. The distribution of these sample plots is shown in figure 2. The survey items included longitude, latitude, altitude, slope degree, age, and diameter at breast height (DBH). The AGC (Mg C ha^{-1}) of each sample plot was calculated as the sum of the total dry AGC of bamboo tree (kg), which was estimated using the Weibull distribution model as expressed in equation (5). This model has an R^2 value of 0.937 and a predictive accuracy of 96.43% (Zhou 2006).

$$\mathrm{AGC} = \left[747.787(\mathrm{DBH})^{2.771}\left\{\frac{0.148(\mathrm{Age})}{0.028+(\mathrm{Age})}\right\}^{5.555} + 3.772\right] \times 0.5042 \qquad (5)$$

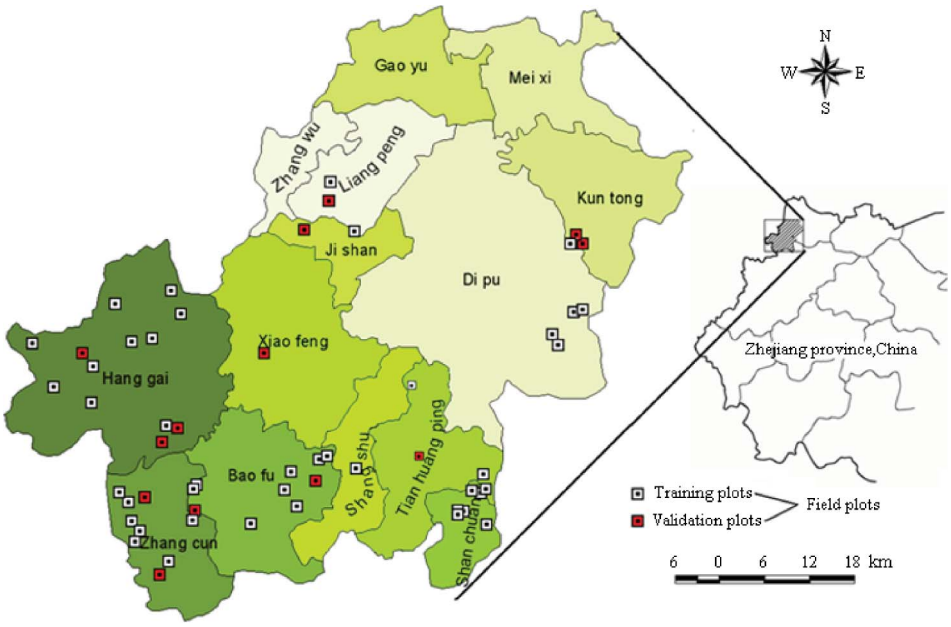

Figure 2. The study area, Anji county, northwest of Zhejiang Province, China, and the spatial distribution of measured sample plots.

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where DBH is measured in cm and Age is in du, the unit used to denote the age of *Moso* bamboo: new bamboo or 1-year bamboo is referred to as 1 du, 2–3 years as 2 du, and 4–5 years as 3 du, and so on. The expansion factor for the conversion from biomass to carbon for *Moso* bamboo forest is equal to 0.5042 (Zhou 2006).

According to statistical analysis of these sample plots, AGC ranges from 9.90 to 34.16 Mg C ha^{-1}. In this study, 41 sample plots were selected randomly and used for developing estimation models and the remaining 13 sample plots were used for model validation.

3.3 *Landsat Thematic Mapper (TM) image processing*

The Landsat TM image (World Reference System, WRS = 119/039) acquired on 5 July 2008 was used in this study. This TM image was rectified to the Transverse Mercator coordinate system by using control points taken from topographic maps, with a root mean square error (RMSE) of 0.51 pixels. The nearest-neighbour algorithm was used to resample the TM image into a pixel size of 30 m $\times$ 30 m during the image rectification. The Dark Object Subtraction (DOS)-based model (Chavez 1996, Song *et al.* 2001, Schroeder *et al.* 2006, Vicente-Serrano *et al.* 2008) was used to conduct radiometric and atmospheric calibration for the TM image. A topographic correction was not applied because the TM image used in this study was little affected by topography.

The atmospherically calibrated TM spectral bands were used to produce many new images, as shown in table 1, summarized according to the following three categories: (1) vegetation indices (Jensen 2005); (2) linear transform; and (3) texture measures (Lu and Batistella 2005). In this study, the co-occurrence measures combining different spectral bands (i.e. bands TM2, 3, 4, 5 and 7) and moving window

Table 1. Image variables used in the study.

Variables	Formula	
Vegetation indices		
DVI	$\mathrm{NIR}_{\mathrm{TM4}} - \mathrm{red}_{\mathrm{TM3}}$	
SR	$\mathrm{NIR}_{\mathrm{TM4}}/\mathrm{red}_{\mathrm{TM3}}$	
NDVI	$(\mathrm{NIR}_{\mathrm{TM\ 4}} - \mathrm{red}_{\mathrm{TM\ 3}})/(\mathrm{NIR}_{\mathrm{TM\ 4}} + \mathrm{red}_{\mathrm{TM\ 3}})$	
TVI	$\sqrt{\mathrm{NDVI} + 0.5}$	
PVI	$(\mathrm{NIR}_{\mathrm{TM4}} - a \times \mathrm{red}_{\mathrm{TM3}} - b)/\sqrt{1+a^2}, a = 1.0902, b = -0.0044$	
IIVI	$(\mathrm{NIR}_{\mathrm{TM4}} - \mathrm{MidIR}_{\mathrm{TM5}})/(\mathrm{NIR}_{\mathrm{TM4}} + \mathrm{MidIR}_{\mathrm{TM5}})$	
SAVI	$1.5 \times (\mathrm{NIR}_{\mathrm{TM4}} - \mathrm{red}_{\mathrm{TM3}})/(\mathrm{NIR}_{\mathrm{TM4}} + \mathrm{red}_{\mathrm{TM3}} + 0.5)$	
EVI	$5 \times (\mathrm{NIR}_{\mathrm{TM4}} - \mathrm{red}_{\mathrm{TM3}})/(\mathrm{NIR}_{\mathrm{TM4}} + 6\mathrm{red}_{\mathrm{TM3}} - 7.5\mathrm{blue}_{\mathrm{TM1}} + 1)$	
Linear transform		
TCT1	$0.291\mathrm{TM1} + 0.249\mathrm{TM2} + 0.481\mathrm{TM3} + 0.557\mathrm{TM4} + 0.444\mathrm{TM5} + 0.171\mathrm{TM7}$	
TCT2	$-0.273\mathrm{TM1} - 0.217\mathrm{TM2} - 0.551\mathrm{TM3} + 0.722\mathrm{TM4} + 0.073\mathrm{TM5} - 0.165\mathrm{TM7}$	
TCT3	$0.145\mathrm{TM1} + 0.176\mathrm{TM2} + 0.332\mathrm{TM3} + 0.340\mathrm{TM4} - 0.621\mathrm{TM5} - 0.419\mathrm{TM7}$	
PCA1	$0.111\mathrm{TM1} - 0.274\mathrm{TM2} + 0.234\mathrm{TM3} - 0.317\mathrm{TM4} + 0.758\mathrm{TM5} - 0.428\mathrm{TM7}$	
PCA2	$0.159\mathrm{TM1} - 0.388\mathrm{TM2} + 0.414\mathrm{TM3} + 0.129\mathrm{TM4} + 0.198\mathrm{TM5} + 0.773\mathrm{TM7}$	
PCA3	$0.152\mathrm{TM1} - 0.537\mathrm{TM2} + 0.432\mathrm{TM3} + 0.230\mathrm{TM4} - 0.503\mathrm{TM5} - 0.442\mathrm{TM7}$	
Texture measures		
Mean	$\sum_{i=0}^{N-1}\sum_{j=0}^{N-1} ip(i,j)$	Note: $p(i,j) = V(i,j)/\sum_{i=0}^{N-1}\sum_{j=0}^{N-1} V(i,j)$, where $V(i,j)$ is the value in the cell i,j (row i and column j) of the moving window and N is the number of rows or columns
Variance	$\sum_{i=0}^{N-1}\sum_{j=0}^{N-1} (i - \mathrm{mean})^2 p(i,\ j)$	$\mu_x = \sum_{i=0}^{N-1} j \sum_{j=0}^{N-1} p(i,j)$
Homogeneity	$\sum_{i=0}^{N-1}\sum_{j=0}^{N-1} \frac{p(i,j)}{1+(i-j)^2}$	$\mu_y = \sum_{i=0}^{N-1} i \sum_{j=0}^{N-1} p(i,j)$
Contrast	$\sum_{\lvert i-j\rvert=0}^{N-1} \lvert i-j\rvert^2 \left\{\sum_{i=1}^{N}\sum_{j=1}^{N} p(i,j)\right\}$	$\sigma_x = \sum_{j=0}^{N-1} (j-\mu_j)^2 \sum_{i=0}^{N-1} p(i,j)$
Dissimilarity	$\sum_{\lvert i-j\rvert=0}^{N-1} \lvert i-j\rvert \left\{\sum_{i=1}^{N}\sum_{j=1}^{N} p(i,j)\right\}$	$\sigma_y = \sum_{i=0}^{N-1} (i-\mu_i)^2 \sum_{j=0}^{N-1} p(i,j)$
Entropy	$-\sum_{i=0}^{N-1}\sum_{j=0}^{N-1} p(i,j)\log(p(i,j))$	
Angular second moment	$\sum_{i=0}^{N-1}\sum_{j=0}^{N-1} p(i,j)^2$	
Correlation	$\frac{\sum_{i=0}^{N-1}\sum_{j=0}^{N-1} (i \cdot j)p(i,j) - \mu_x\mu_y}{\sigma_x\sigma_y}$	

Note: DVI – Difference Vegetation Index; SR – Simple Ratio; NDVI – Normalized Difference Vegetation Index; TVI –Triangular Vegetation Index; PVI: Perpendicular Vegetation Index; IIVI – Infrared Index Vegetation Index; SAVI – Soil Adjusted Vegetation Index; EVI – Enhanced Vegetation Index; TCT – Tasselled Cap Transformation; PCA – Principal Component Analysis.

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sizes (i.e. 5 × 5, 7 × 7, 9 × 9, 11 × 11, 15 × 15, 19 × 19 and 25 × 25 pixels) were examined. Correlation analysis was used to select potential textural images for the AGC modelling.

3.4 *Model development*

Based on their coordinates, the sample plots were linked to Landsat TM-derived variables such as spectral bands, vegetation indices, and textural images. To reduce some of the errors of image analysis caused by errors in the image registration and sample plot positions, a window size of 3 × 3 pixels was used to extract the mean values of the selected remote sensing variables for each plot. These samples were used to examine the relationships between variables and AGC of *Moso* bamboo forests, and then used to establish carbon estimation models.

A regression model was developed and stepwise selection was performed to select variables for the model, defining $\alpha = 0.1$ and $\beta = 0.15$ as thresholds for adding and removing an independent variable, respectively. Multicollinearity issues were addressed by calculating and monitoring the size of the condition index. The model with the condition index value greater than 30 was not accepted, indicating that there was no serious collinearity inherent in the selected models (Weisberg 1985, Næsset *et al.* 2005).

Estimation with PLS typically follows a sequential procedure with user's interaction and decisions during the process (Næsset *et al.* 2005). In this study, a principal component analysis was first implemented, followed by an initial estimation of all models. The method for choosing the optimal number of components was described in §2.1. Meanwhile, the PLS-Bootstrap algorithm was used to select variables; details of the algorithm are presented in Lazraq *et al.* (2003) and Wang *et al.* (2006).

The variables selected by the PLS-Bootstrap algorithm were used as input variables of the Erf-BP networks, which were the same as those used in the PLS model. There is a lack of scientific methods to determine the structure of a neural network, such as training performance and the number of neurons in hidden layers. In this study, based on a trial-and-error procedure, the optimized neural network structure was determined when the sum of the MREs of the training set and of validation set was minimized. The hidden neurons were set in a range of [7, 13] with an increased step of 1. The training performance was set in a range of [0.2, 0.1] with a decreased step of 0.01. The number of hidden layers was set to one. The performance function for the networks was the sum square error (SSE). The value of the learning rate was set to 0.2. The momentum factor was set to 0.9, and λ was set to one. The input (equation (6)) and output (equation (7)) data were normalized as follows:

$$X_{\text{norm}} = 2 \times \frac{X - X_{\min}}{X_{\max} - X_{\min}} - 1 \tag{6}$$

$$Y_{\text{norm}} = \frac{Y - Y_{\min}}{Y_{\max} - Y_{\min}} \tag{7}$$

where, respectively, X and Y are the original input and output data, X_{norm} and Y_{norm} are the normalized values ($X_{\text{norm}} \in [-1, 1]$, $Y_{\text{norm}} \in [0, 1]$), $X_{\min}$ and $X_{\max}$ are the minimum and maximum values of the input data, and $Y_{\min}$ and $Y_{\max}$ are the minimum and maximum values of the output data.

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3.5 *Model evaluation and comparison*

Assessment of AGC at a per-pixel level is often difficult, and the accuracy may be misleading due to the registration errors between sample plots and image (Lu 2006). Therefore, the estimated results were evaluated based on a window size of 3 × 3 pixels. The reliability of the models were measured by the correlation coefficient (r), MRE, RMSE (Mg C ha^{-1}) and relative RMSE ($RMSE_r$, which is equal to the RMSE divided by the mean of the measured AGC × 100%) (Muukkonen and Heiskanen 2005). The total and average AGC of *Moso* bamboo estimated from the linear regression, PLS and Erf-BP models for the entire county were also compared with the estimated results from the traditional inventory method (equations (8) and (9)).

$$AGC_{average} = (ASD) \times (AM) \tag{8}$$

$$AGC_{total} = (AGC_{average}) \times A \tag{9}$$

where AGC_{total} and $AGC_{average}$ are the total and average AGC of *Moso* bamboo per unit area, respectively (in Mg C ha^{-1}); ASD is the average stem density of *Moso* bamboo per unit area (in culms ha^{-1}); AM is the average mass of AGC per culm (in kg C culm^{-1}); and A is the total area of *Moso* bamboo (in ha). In this study, ASD was estimated as 3225 culms ha^{-1} based on the 54 sample plots, AM was 6.45 kg C culm^{-1} (Zhou 2006), and A was 56 872.73 ha calculated by the classification map.

4. Results

4.1 *Analysis of Landsat TM-derived variables*

Pearson's correlation coefficients between the independent variables and AGC of *Moso* bamboo (see table 2) indicate that VariB7_5 has the highest r value, followed by CorrB4_9, and band 4. The spectral reflectance bands show a significant correlation with AGC, except for band 5. For the vegetation indices, AGC is significantly correlated with DVI, PVI, IIVI and SAVI, but is weakly related to NDVI, SR and TVI. The linearly transformed images (i.e. PCA1, PCA3, TCT 1 and TCT 2) are significantly correlated with AGC. Most of textural images are weakly correlated with AGC, but some textural images, which have an optimal combination of texture measure, spectral band and window size, are significantly correlated with AGC, as

Table 2. Pearson's correlation coefficients between independent variables and AGC.

Variable	r	Variable	r	Variable	r	Variable	r
Band 2	0.338*	SR	–0.007	PCA2	0.123	CorrB4_9	–0.362**
Band 3	0.293*	TVI	–0.056	PCA3	0.313*	ContB4_11	0.290*
Band 4	0.355**	PVI	0.316*	TCT1	0.341*	MeanB2_25	0.286*
Band 5	0.217	IIVI	0.321*	TCT2	0.298*	MeanB4_25	0.257
Band 7	0.333*	SAVI	0.275*	TCT3	0.167	MeanB7_25	0.207
NDVI	–0.053	EVI	0.255	VariB7_5	0.377**	AsmB2_25	–0.244
DVI	0.320*	PCA1	0.338*	CorrB3_7	–0.314*	EntrB3_25	0.267

Level of significance, *< 0.05, **< 0.01; VariBa_b indicates the variance texture measure with the TM band a and associated window size of $b \times b$ pixels. Corr, Cont, Mean, Asm, and Entr represent correlation, contrast, mean, angular second moment, and entropy texture measures, respectively.

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summarized in table 2. For example, the textural image based on the variance texture measure with TM band 7 and window size of 5 × 5 pixels (i.e. VariB7_5) or the textural image based on the correlation texture measure with TM band 4 and window size of 9 × 9 pixels (CorrB4_9) have higher correlation coefficients than any other spectral bands or vegetation indices.

4.2 *Comparison and evaluation of model performance*

Initial TM-derived variables were selected (see table 2) based on the correlation analysis whose variables had a relatively good correlation with AGC. A stepwise regression analysis was then used to establish the linear regression model as expressed in equation (10). CorrB4_9 and VariB7_5 were selected for input into the linear regression model.

$$\mathrm{AGC_{linear}} = 18.445 - 0.140(\mathrm{CorrB4_9}) + 1.279(\mathrm{VariB7_5}) \quad R^2 = 0.247 \tag{10}$$

The correlation analysis shows significant relationships between AGC and CorrB4_9 and VariB7_5 (see table 2). According to the *F*-test, the developed linear regression model is significant at a 0.005 level. The *t*-test indicates that both selected variables in this model are significant at 0.044 (for CorrB4_9) and 0.055 (for VariB7_5) levels.

As described in §3.4, the PLS-Bootstrap algorithm was used to identify the variables used in the PLS model. The variables of band 5, band 7, DVI and IIVI were selected from those variables given in table 2 and then used to build the PLS model. Thus, the optimal model was obtained when the optimal number of PLS components is equal to two (see figure 3(*a*)) and is expressed by equation (11):

$$\begin{aligned}\mathrm{AGC_{PLS}} = {} & -22.168 - 81.007(\mathrm{Band5}) + 499.308(\mathrm{Band7}) - 90.636(\mathrm{DVI}) \\ & + 156.87(\mathrm{IIVI})\end{aligned} \tag{11}$$

The variables used to train the Erf-BP model were the same as those used in the PLS model. According to the experimental design described in §3.4, the optimal neural network structure was obtained (see figure 3(*b*)). Figure 3(*b*) shows that the optimal hidden neurons number and training performance were 9 and 0.11, respectively. Thus, the Erf-BP-based AGC prediction model is expressed in equation (12):

$$\mathrm{AGC_{Erf-BP}} = \frac{Y_{\max} - Y_{\min}}{(1 + \exp(-\mathrm{Erf}(X \times \mathrm{IW} + \mathrm{ones}(n,1) \times b_1) \times \mathrm{LW} + \mathrm{ones}(n,1) \times b_2)))} + Y_{\min} \tag{12}$$

where $Y_{\min}$ and $Y_{\max}$ are the minimum and maximum values of the measured AGC, respectively, X is the input sample, IW and LW are the connection weights from the input layer to the hidden layer and from the hidden layer to the output layer, respectively, b_1 and b_2 are the threshold values of the hidden layer and output layer, respectively, and ones(n,1) is an n-by-1 matrix of ones.

A comparative analysis of the predictive performance of the three different models indicated that the Erf-BP model provided the best estimation performance (see table 3). Based on the training or validation data sets, the Erf-BP model produces the highest r and lowest MRE, RMSE and $\mathrm{RMSE_r}$ compared with the other two models. The different predictive performances of these three models are also shown in figure 4. In the linear regression model, the slope of the measured versus the predicted AGC is near

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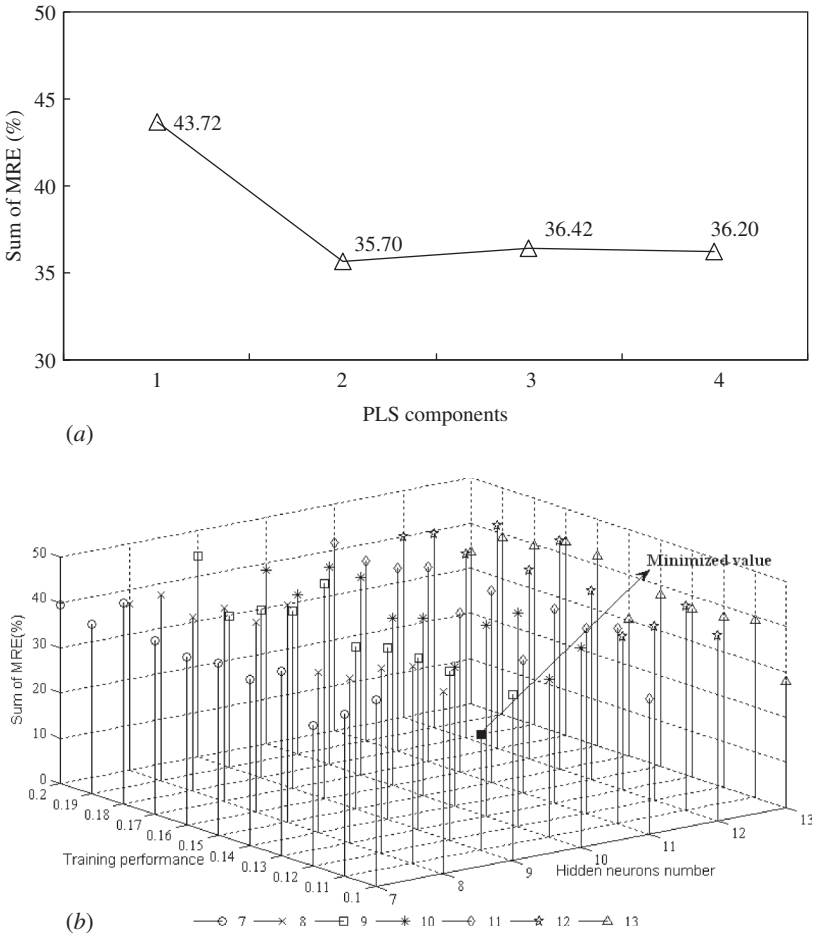

Figure 3. The sum of MREs of training and validation sets with (*a*) different PLS components and (*b*) different neural network structures.

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zero, and the intercept is near the average of the measured AGC (see figure 4(*a*)). This implies that the best-fit line of the linear regression model approaches a horizontal line and its prediction capability is fairly poor. In comparison with the linear regression, the PLS model improves the estimation performance (see figure 4(*b*)), but both the linear

Table 3. Reliability of the linear regression, PLS and Erf-BP models.

	Training set				Validation set			
Model	r	MRE (%)	RMSE (Mg C ha^{-1})	RMSE$_r$ (%)	r	MRE (%)	RMSE (Mg C ha^{-1})	RMSE$_r$ (%)
Linear	0.497	21.80	4.848	23.30	–0.055	22.46	5.024	25.89
PLS	0.581	18.94	4.545	21.84	0.573	16.76	4.065	20.95
Erf-BP	0.955	8.10	1.998	9.60	0.761	17.42	3.969	20.46

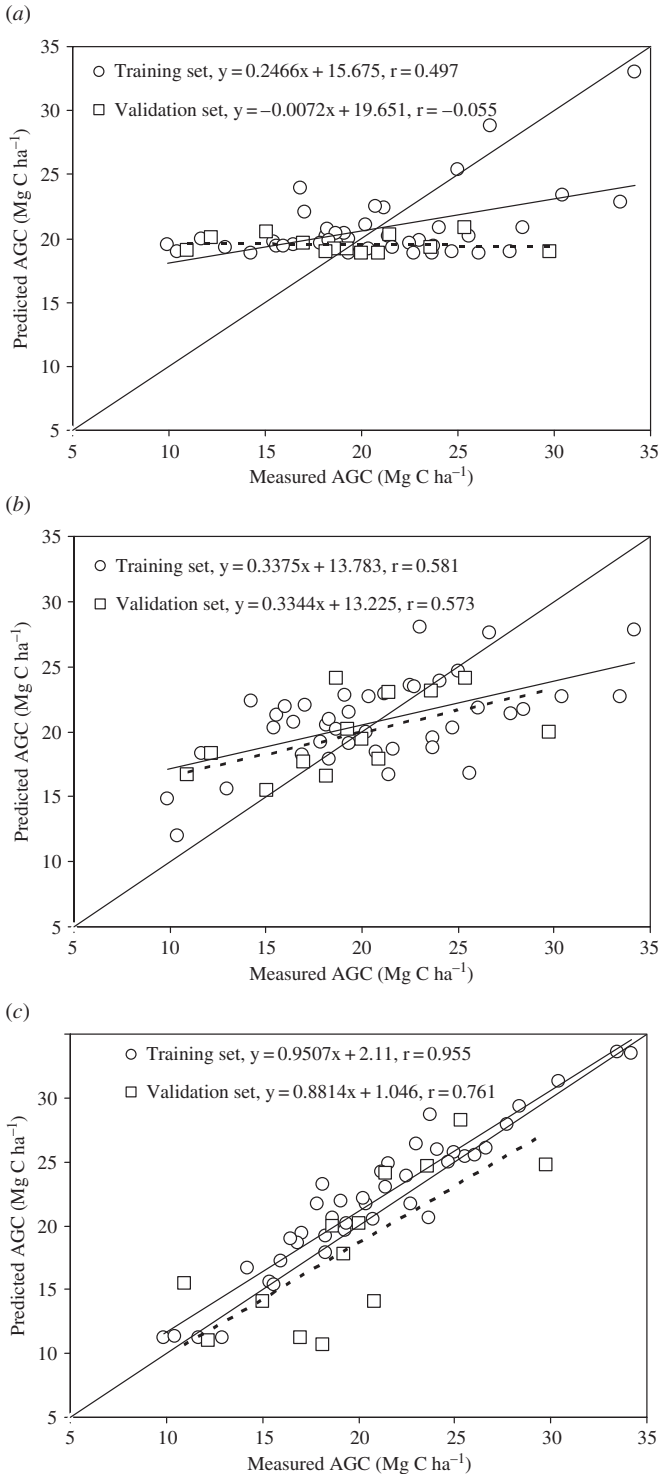

Figure 4. A comparison of the relationship between predicted AGC versus the measured AGC based on different models: (*a*) the linear regression model, (*b*) the PLS model, and (*c*) the Erf-BP model.

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Table 4. Comparison of estimated AGC with different methods.

Method	Total AGC (Tg C)	Average AGC (Mg C ha^{-1})
Linear	1.206	21.20
PLS	1.174	20.64
Erf-BP	1.184	20.82
Traditional inventory	1.183	20.80

Note: 1Tg = 10^{12}g.

regression and PLS models have similar problems in overestimating areas with low AGC and underestimating areas with high AGC. The estimation performance of the Erf-BP model is more reliable in comparison with both the linear and the PLS models (see figure 4).

An alternative to evaluating model performance is to compare the total AGC and average AGC values, as summarized in table 4. Both the Erf-BP model and the traditional inventory method have nearly the same AGC, between the highest value from the linear regression model and the lowest value from the PLS model. The results in table 4 are consistent with results shown in table 3 and figure 4. According to these results, the Erf-BP model performs the best, while the PLS and linear regression models perform poorly. These results indicate that the Erf-BP model is a suitable method for estimating the AGC of *Moso* bamboo in a large area.

The three models developed were used to estimate AGC for the entire study area, and their spatial distributions are shown in figure 5. The histograms of the AGC values in figure 5 show that most of the *Moso* bamboo in the forested area has AGC values between 10 and 30 Mg C ha^{-1}. A detailed examination of the AGC maps and the histograms of AGC values reveal AGC differences among the three AGC maps. Most of the AGC values estimated by the linear regression and PLS models have data ranging between 15 and 25 Mg C ha^{-1}, and few are in the low and high AGC levels (see figures 5(*a*) and 5(*b*)). The low and high AGC levels are better represented by the Erf-BP model than the other two models (see figures 4(*c*) and 5(*c*)). The linear regression and PLS models overestimate and underestimate the AGC levels, respectively, and produce some extreme values, for example negative AGC values, which are consistently set to zero (Fuchs *et al.* 2009). In this study, the ratio of extreme values to all AGC values is small. The mean and total AGC are well preserved, which corresponds to the findings of Labrecque *et al.* (2006) and Fuchs *et al.* (2009).

5. Discussion

Many factors can affect the estimation performance with remote sensing-based methods (Tomppo *et al.* 2002). Attention should be paid to the quality of samples, geometric accuracy between Landsat TM image and sample plots, selection of suitable remote sensing variables, and use of suitable algorithms (Lu 2006). First, collection of sufficient sample plots with good representativeness is crucial for examining the relationships between AGC and remote sensing variables in developing AGC estimation models (Tomppo *et al.* 2002, Fuchs *et al.* 2009). It is important to make sure that the collected samples are from different stand structures, management measures and geographic conditions, and have samples from both low and high AGC levels. Second, accurate geometric registration between the Landsat TM data and sample plots is required. Because of the image rectification error, the Global Positioning System (GPS) location

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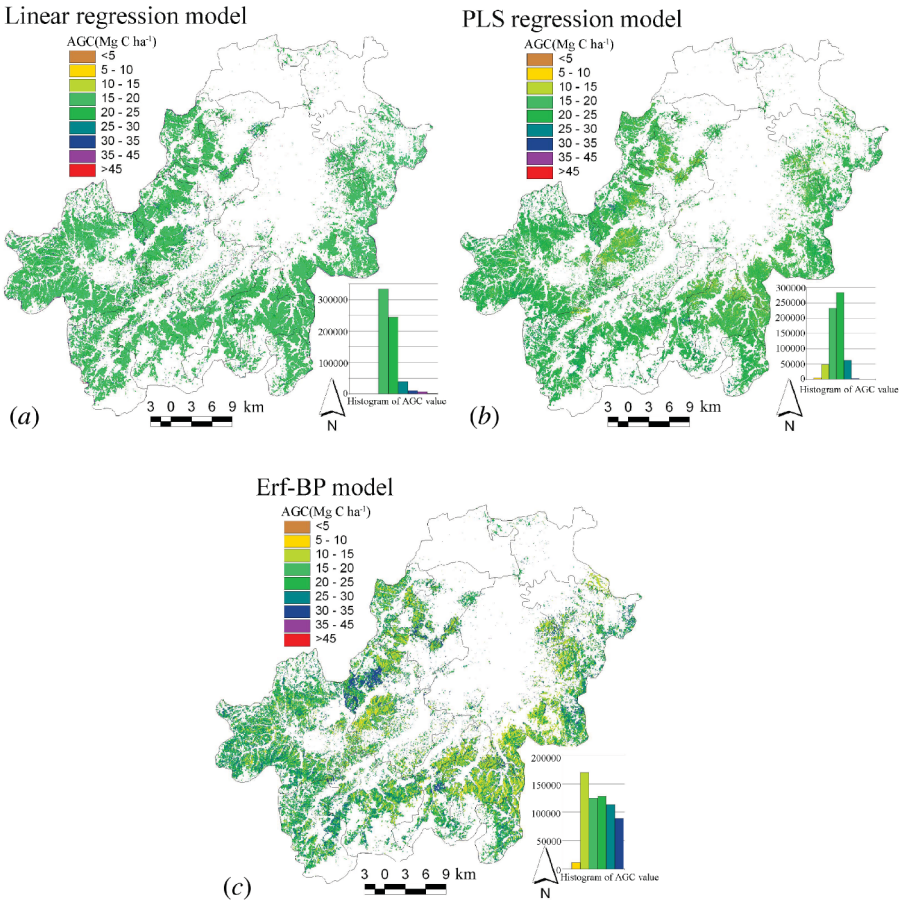

Figure 5. Comparison of AGC spatial distribution for Anji county, China, estimated using: (*a*) the linear regression model, (*b*) the PLS model, and (*c*) the Erf-BP model.

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error and the constraint of plot size, it is difficult to make sure that the sample plot accurately represents a single pixel value on the Landsat TM image. Therefore, a window size of 3 × 3 pixels is used to extract a mean value for each sample plot to reduce the geometric errors between image and sample plots.

In practice, selection of suitable variables is one of the most important issues for developing an AGC estimation model and contributes to a considerable reduction in estimation errors (Lu 2006, Fuchs *et al.* 2009). In this study, one reason why the estimated accuracy of PLS was better than the linear regression model is that the PLS-Bootstrap algorithm was used to select the variables. The selection of variables using stepwise regression analysis is simply based on the relationships between AGC and variables. Because the relationships between AGC of *Moso* bamboo and variables are weak (see table 2), it is necessary to set suitable thresholds for adding and removing a variable. In this study, CorrB4_9 and VariB7_5 were selected as variables for input into the linear regression model, but the estimated results were poor (see table 3 and figure 4(*a*)). The main reason is that the stepwise regression analysis

cannot effectively handle the multicollinearity problem (De Jong *et al.* 2003, Chong and Jun 2005, Ju *et al.* 2008).

However, the PLS-Bootstrap algorithm is recommended for selecting suitable variables for various reasons. First, PLS regression has been proposed as a technique to find latent structures in a large number of variables by reducing the number of variables to a few non-correlated principal components (Næsset *et al.* 2005). PLS regression also has a modelling advantage in dealing with multicollinearity among variables and using explanatory information inherent in variables whose relationships with AGC are insignificant (Chong and Jun 2005, Næsset *et al.* 2005, Cho *et al.* 2007, Ju *et al.* 2008, Wolter *et al.* 2008). For example, although there exists serious multicollinearity between the variables of bands 5 and 7 ($r = 0.933$), the variable of band 5 is insignificant (see table 2). In this study, the variables of bands 5 and 7 were both selected for input into the PLS model (equation (11)), and the estimated results of the PLS model were improved (see figure 4(*b*)). Second, the bootstrap method is a collection of sample reuse techniques designed to estimate a confident threshold of regression coefficients, which is a preferable sampling method (Efron and Tibshirani 1993, Amato and Vinzi 2003, Wisnowski *et al.* 2003). Third, the variables selected by the PLS-Bootstrap algorithm are very stable in the sense that the algorithm is repeated many times (Lazraq *et al.* 2003). In this study, the PLS-Bootstrap algorithm was repeated 200 times.

In reality, the relationships between AGC and remote sensing-derived variables are not linear. Both the linear regression and PLS models cannot effectively deal with nonlinear relationships. BP-ANN as a nonparametric technique can approximate any arbitrarily complex nonlinear function regardless of the complexity of the relationships between the variables and AGC, and has the advantage of fault tolerance, which makes it insensitive to noise. The most significant advantage of the Erf-BP model is its ability to estimate extreme values of AGC, which is not done as efficiently by the PLS model, and appears to be a weakness for the linear regression (see figures 4 and 5). The AGC range estimated by remote sensing using the Erf-BP model is consistent with the plot of AGC ranging from 9.90 to 34.16 Mg C ha^{-1} in this study, and is also consistent with the results (12.60–36.80 Mg C ha^{-1}) reported by Uchimura (1980). The total and average AGC estimated by remote sensing using the three methods was similar to the results of the traditional inventory method (see table 4). Comparing the estimated AGC of *Moso* bamboo based on remote sensing with the traditional inventory method, this study indicates that remote sensing is an effective method that can be used to estimate AGC of *Moso* bamboo.

Although the model performances of the PLS regression and Erf-BP model are preferable, they also have some drawbacks. One problem, for example, is that both methods need to determine the parameters that affect the estimation results significantly (see figure 3). It is time-consuming to choose suitable parameters. The adjustment of each parameter to optimize the whole network is an overall challenging but important task (Lee 2004).

6. Conclusions

This study has examined the integration of Landsat TM and field measurement for AGC estimation through a comprehensive analysis of different models. The results show that the remote sensing technique is a promising tool for estimation of AGC stock of *Moso* bamboo in large areas. The prediction accuracy of the PLS and Erf-BP

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models is better than that of the linear regression model. AGC is overestimated at low AGC levels and underestimated at high AGC levels when the linear regression and PLS models are used. On the contrary, extreme values of AGC are estimated particularly well by the Erf-BP model, which is recommended for *Moso* bamboo's AGC estimation in a large area.

Acknowledgements

We acknowledge support from the National Natural Science Foundation (grant nos 30700638 and 30771715), the '948' item of the National Forestry Bureau (grant no. 2008-4-49), the national '863' program (grant no. 2006AA12Z104) and items of the Science and Technology Department of Zhejiang Province (grant nos 2007C13041 and 2008C12068) and thank the Anji Forestry Bureau for assistance during the field inventory. We also thank the reviewers for their constructive comments and assistance in improving this paper.

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